

# SMART ATTENDANCE SYSTEM

DEVELOPED BY:

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AI-Powered Biometric Classroom Attendance Tracking (BIT4543)

## THE CHALLENGE

### MANUAL ATTENDANCE FAILS

- Time Lost** (5-10 mins/class)
- Proxy Attendance** (Students signing-in for absent peers)
- Large Halls** (Massive classrooms scale chaos)
- Error-Prone** (Missing names & manual entry typos)

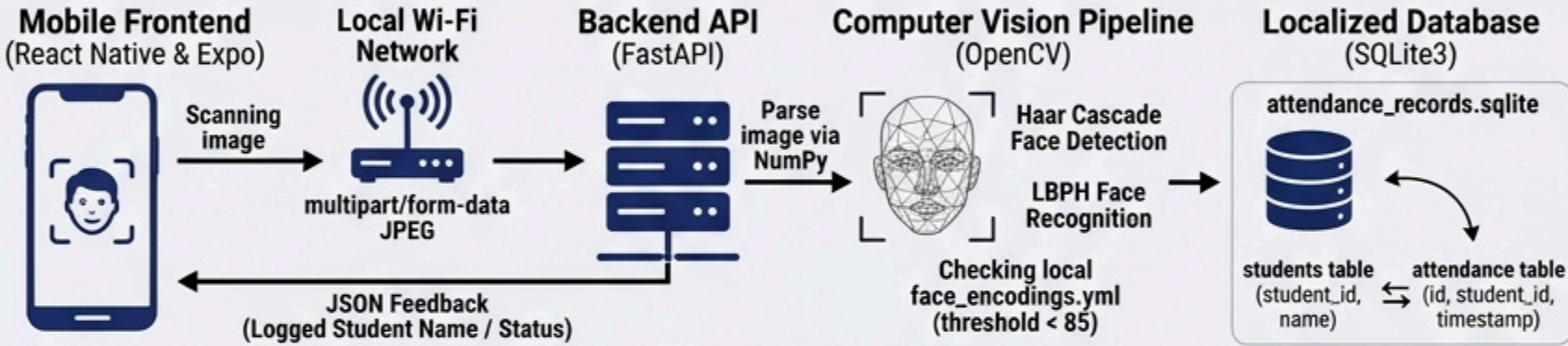
VS.

## THE SOLUTION

### AUTOMATED BIOMETRIC RECOGNITION

- 100% Biometric** (Facial recognition validates physical presence)
- Touchless** (Contactless non-intrusive scan)
- Automated** (Automatic timestamp logs straight to SQLite)
- Proxy-Proof** (Tamper-proof, blocks peer sign-ins)

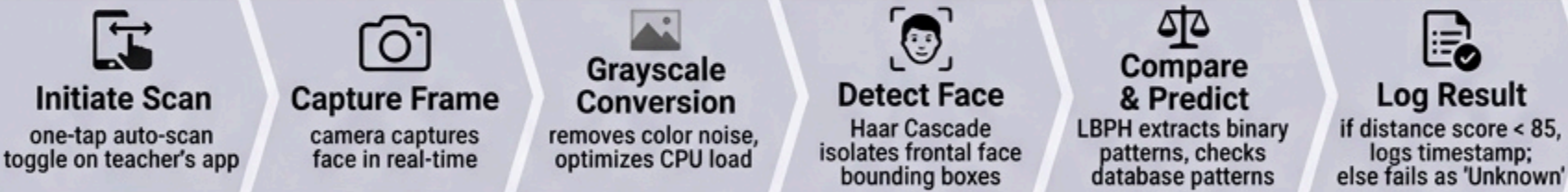
## SYSTEM & SERVER ARCHITECTURE



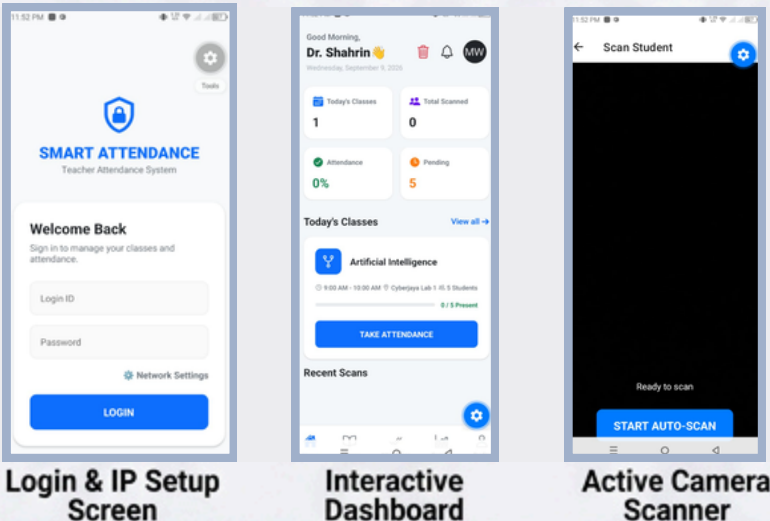
## 100% OFFLINE CAPABILITY

No cloud APIs or internet required during scans, protects biometric privacy, zero cost/maintenance, runs on a local standard router.

## 6-Step Recognition Pipeline

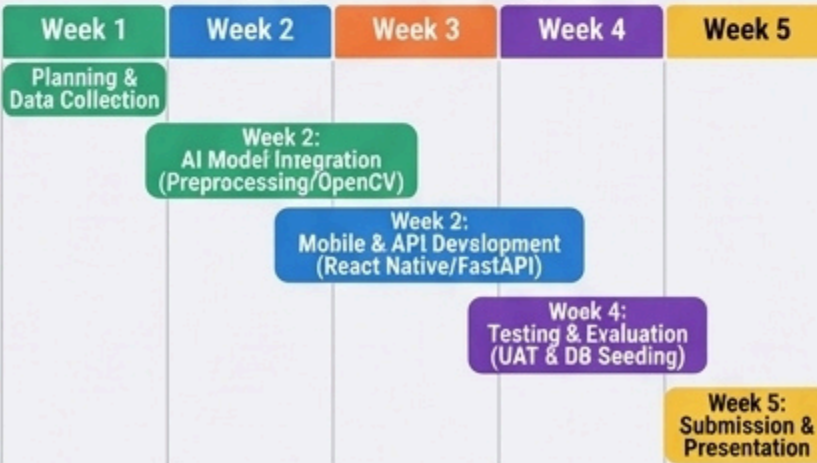


## Educator-Centric User Interface



## Project Timeline & Delivery

(adapted from <image 1>)



**Key Takeaways:** From roll call to smart attendance. Lightweight computer vision automates record-keeping and secures data integrity.

